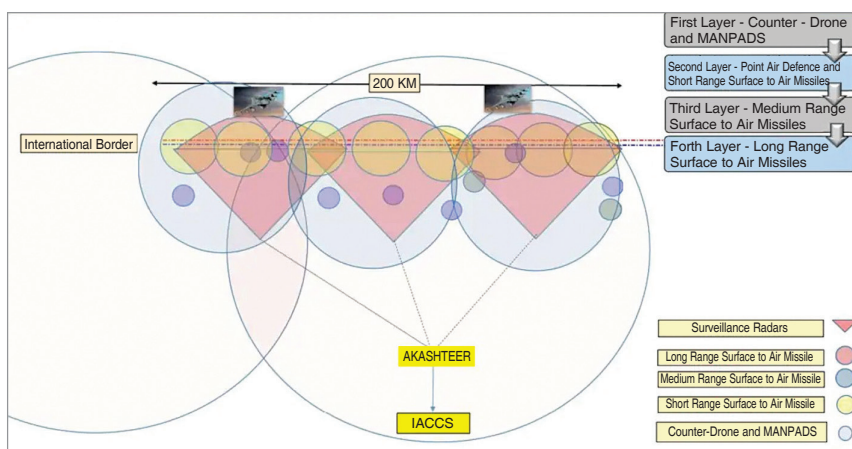




The integrated air command and control system (IACCS) serves as an anchor
(Source: <https://www.ndtv.com>)

Box 2: Interceptor system

The interceptor system is a tiered defensive shield that integrates long-, medium-, and short-range missile systems into a unified architecture for comprehensive target engagement. In the outermost tier, long-range interceptors provide coverage against high-altitude and distant threats. The medium-range tier adds redundancy against manoeuvrable targets such as cruise missiles, aircraft, and drones. The inner tier comprises interceptors designed to engage short-range targets such as rockets, artillery shells, low-flying drones, and precision-guided munitions.

platforms into a unified framework. Operational protocols are aligned and standardised across services, while baseline capabilities such as secure data exchange, coordinated detection, and initial interceptor deployment are validated. Pilot deployments, communication resilience testing, and controlled exercises are conducted to confirm that the integrated system operates reliably under multi-domain conditions.

The outcome is a robust, secure, and indigenous communication backbone capable of supporting national defence operations.

Phase II: System expansion and strategic operational deployment (2030-2035). Building on the foundation laid in Phase I, Phase II advances the system towards full operational deployment across theatres.

Integrated networks are expanded to enable seamless multi-domain operations in coordination with defence and national infrastructure. Cutting-edge technologies are incorporated, including hypersonic weapons, directed-energy platforms, quantum and terahertz

communication systems, AI-enabled autonomous systems, and space-based surveillance assets.

This phase also strengthens resilience against cyber threats, enhances adaptability in contested environments, and deepens strategic autonomy by indigenising critical supply chains while reducing dependence on external sources for sensitive technologies and components.

The outcome is a fully operational, scalable system-of-systems that reinforces technological *Atmanirbharta* (self-reliance).



Russia's S-400 air defence system plays a vital role in Mission Sudarshan Chakra (Source: <https://indianexpress.com>)

Mission Sudarshan Chakra: Global support

Many of the technologies involved are highly complex, requiring specialised expertise and critical components that are often scarce and difficult to source. India has collaborated with multiple countries under various collaborative mechanisms.

Key agreements and MoUs have been signed with the following partner countries:

Russia. India and Russia share a long-standing defence partnership, with several collaborative programmes.

Russia has supplied the S-400 air defence system along with technology transfer. Integrated into the air defence grid, the system enhances long-range detection and interception of high-speed aerial threats.

Agreements with Russia exist for the licensed production of platforms such as the Su-30MKI aircraft and T-90 tanks. Russia has now offered to collaborate on the Su-57E fifth-generation fighter, thereby opening access to stealth and propulsion technologies.

In addition, agreements are at an advanced stage for the joint development of hypersonic propulsion and directed-energy weapon systems.

Israel. India has a number of collaborative programmes with Israel, including the Medium Range Surface-to-Air Missile (MR SAM), the Long-Range Surface-to-Air Missile (LR SAM/Barak 8), and advanced radar